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RESEARCH PROFILE

- Specializing in AI-driven cybersecurity and adaptive Intrusion Detection Systems for Industrial IoT networks.
- Driven to develop edge-tractable, poison-resistant defense frameworks for autonomous zero-day threat mitigation.
- Seeking a postdoctoral position to leverage comprehensive expertise in generative deep learning, game-theoretic optimization, resource-aware deployment, and trustless collaborative networks in real-world security applications.

AREAS OF INTEREST

- Generative Deep Learning
- Spatio-Temporal Graphs
- Continual Openset Learning
- Zero-Day Threat Mitigation
- Industrial IoT Security
- Network Trust Management
- Edge and Fog Computing
- Collaborative Defense
- Game-Theoretic Optimization

ACADEMIC BACKGROUND

- 2021 – Present ➤ **Doctor of Philosophy (Ph.D.) — Computer Science and Engineering.**
1. **Indian Institute of Technology Guwahati, Assam, India**, jointly with
2. **Indian Institute of Technology (BHU) Varanasi, Uttar Pradesh, India.**
CGPA: 8.75/10.00
Thesis title: *Resource-Optimized Mimicry Detection in Industrial Internet of Things Networks Using Deep Learning Techniques.*
- 2019 – 2021 ➤ **Master of Technology (M.Tech.) — Computer Science and Engineering.**
National Institute of Technology Durgapur (NIT Durgapur), West Bengal, India.
CGPA: 9.28/10.00
Thesis title: *A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond.*
- 2006 – 2010 ➤ **Bachelor of Engineering (B.E.) — Computer Science and Engineering.**
University Institute of Technology (UIT), West Bengal, India.
Affiliated with The University of Burdwan (BU).
Result: 79.9%
- 2006 ➤ **CBSE AISSCE (Class XII).**
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Subjects: Physics, Chemistry, Mathematics, Biology, English.
Result: 79.6%
- 2004 ➤ **CBSE AISSE (Class X).**
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Result: 82.8%

COMPETITIVE EXAMS

- TOEFL – 2021 ➤ **Secured 107 marks out of a total 120**, conducted by the *Educational Testing Service (ETS).*
- GATE – 2019 ➤ **Secured 96.88 percentile score** in the subject *Computer Science and Information Technology*, conducted by the *Indian Institute of Technology (IIT) Madras.*

RESEARCH EXPERIENCE

- 2021 – Present
- **Doctor of Philosophy (Ph.D.):** IIT Guwahati and IIT (BHU) Varanasi
Focus: AI-Driven Cybersecurity & Adaptive IDS for IoT Networks
 - Architected spatio-temporal graph diffusion ensembles utilizing GCN-GRU backbones to execute explainable, zero-day mimicry attack detection at the resource-constrained IoT edge.
 - Engineered conjoint detection mechanisms via Multi-resolution Ring Buffers and random convolutional kernel transform (ROCKET) models, fusing host behavior with network context for continuous anomaly fingerprinting and $\mathcal{O}(1)$ swarm vaccinations.
 - Designed lightweight, privacy-preserving collaborative IDSs using Bloom filters and distributed LSTM Autoencoders, ensuring robust consensus and adaptive scheduling even in majority-dishonest peer environments.
 - Developed autonomous, zero-overhead trust management and adaptive placement algorithms, dynamically orchestrating heterogeneous deep learning ensembles (e.g., InceptionTime, Dilated TCN, Time Series Mixer) to maximize network lifetime.
- 2019 – 2021
- **Master of Technology (M.Tech.):** NIT Durgapur
Focus: Game-Theoretic Frameworks & Smart City Optimization
 - Formulated game-theoretic models and incentive mechanisms, balancing exploration and exploitation to optimize volunteer mobility and surplus food distribution in dynamic urban environments, leveraging available smart city platforms.
 - Devised dynamic preference aggregation frameworks using bucketing algorithms and Shapley values to optimize crowdsourced project selection and enterprise decision-making.
 - Enhanced intelligent urban transportation systems by optimizing curvelet entropy features for voice-assisted traffic sign recognition.

RESEARCH PUBLICATIONS

PUBLISHED AND ACCEPTED JOURNAL ARTICLES

1. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). CIPHER: A Collaborative IDS Placement and Scheduling Framework for Host-Oriented Mimicry Attack Detection in IoT Networks. *IEEE Internet of Things Journal*. (IF: 8.9). [doi:10.1109/JIOT.2026.3689273](https://doi.org/10.1109/JIOT.2026.3689273)
2. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). LITMUS: A Lightweight Collaborative Intrusion Detection System for Host Oriented Mimicry Attack Detection in Industrial Internet of Things Networks with Dishonest Collaborators in the Majority. *Computer Networks*, 286, 112474–112492. (IF: 4.7). [doi:10.1016/j.comnet.2026.112474](https://doi.org/10.1016/j.comnet.2026.112474)
3. **Sanyal, S.**, Mukhopadhyay, S., & Xhafa, F. (2023). Enhanced Incentive Schemes for Volunteers to Distribute Surplus Food in Smart Cities. *Sustainable Cities and Society*, 91, 104409. (IF: 13.3). [doi:10.1016/j.scs.2023.104409](https://doi.org/10.1016/j.scs.2023.104409)
4. **Sanyal, S.**, Kumar Singh, V., Xhafa, F., Sanyal, B., & Mukhopadhyay, S. (2021). A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond. *Applied Sciences*, 11(11), 5058. (IF: 2.9). [doi:10.3390/app11115058](https://doi.org/10.3390/app11115058)

JOURNAL ARTICLES UNDER REVIEW AND REVISION

1. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). MIRAGE: An Adaptive Open-Set Spatio-Temporal Graph Diffusion Ensemble for Explainable Zero-Day Mimicry Detection at the IoT Edge. *Expert Systems With Applications*, 1–42.
2. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). HADRON: Autonomous Placement and Scheduling of Heterogeneous IDS using Silent Trust for Mimicry Attack Detection in IoT Networks. *Computer Networks*, 1–19.
3. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). JARVIS: Ultralight Conjoint Mimicry Attack Detection with Continuous Anomaly Fingerprinting and Fast Swarm Vaccinations in IoT Networks. *Internet of Things*, 1–24.

4. **Sanyal, S.**, Mukhopadhyay, S., Sanyal, B., & Mukhopadhyay, J. (2026). A Dynamic Preference Aggregation Mechanism using Shapley Scores and K-Means Clustering for Enterprise Decision Making in Smart Cities. *Knowledge and Information Systems*, 1–26.
5. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). SIPHON: A Parent Switch Indifferent Placement and Scheduling of Lightweight IDS for Mimicry Attack Detection in IoT Networks. *IEEE Transactions on Reliability*, 1–12.
6. Kumar, R., **Sanyal, S.**, Mukhopadhyay, S., & Xhafa, F. (2025). Volunteer Mobility Management for Surplus Food Distribution in Dynamic Environments in Smart Cities. *Sustainable Cities and Society*, 1–94. **(Under Revision)**.

CONFERENCE PROCEEDINGS

1. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, July). A Survey of Project Selection in Tourist Spots using Smart City Platforms. In *Proc. of the 20th International Conference on Complex, Intelligent, and Software Intensive Systems (CISIS-2026)* (pp. 1–12). Springer. [doi:10.1007/978-3-032-29430-2_11](https://doi.org/10.1007/978-3-032-29430-2_11)
2. Sanyal, B., & **Sanyal, S.** (2026, June). Traffic Sign Recognition with Voice Assistance Using Optimized Curvelet Entropy Features. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11648793](https://doi.org/10.1109/GCON69192.2026.11648793)
3. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026, June). CSIDS: Detecting Mimicry Attacks using Network Context with Host Behavior in IoT Networks. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11649178](https://doi.org/10.1109/GCON69192.2026.11649178)
4. **Sanyal, S.**, Priya, T., Debnath, N., Mitra, D., & Mukhopadhyay, S. (2026, June). Placing Tourist Spots in Smart Cities: A Crowdsourced Game Theoretic Approach. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11648713](https://doi.org/10.1109/GCON69192.2026.11648713)
5. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, May). FIREBOLT: Visitor Frequency Aware Crowdsourced Project Selection for Tourist Spots in Smart Cities using Bucketed Shapley Scores. In *Proc. of the 14th International Conference on Emerging Internet, Data & Web Technologies (EIDWT-2026)* (Vol. 291, pp. 255–266). Springer. [doi:10.1007/978-3-032-22798-0_23](https://doi.org/10.1007/978-3-032-22798-0_23)
6. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2024, December). An Adaptive IDS Scheduling Framework for Host Based Mimicry Attack Detection in IoT Networks. In *Proc. of 2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS 2024)* (pp. 1–6). IEEE. [doi:10.1109/ANTS63515.2024.10898203](https://doi.org/10.1109/ANTS63515.2024.10898203)
7. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2024, June). Crowdsourced Project Selection in Tourist Spots with Voting in Smart Cities. In *Proc. of the Decision Science Alliance International Summer Conference (DSA ISC 2024)* (pp. 105–119). Springer. [doi:10.1007/978-3-031-78241-1_10](https://doi.org/10.1007/978-3-031-78241-1_10)
8. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., Singh, V. K., & Sanyal, B. (2022, August). An Incentive Framework for Surplus Food Distribution in the Context of Smart Cities. In *Proc. of the International Conference on Network-Based Information Systems (NBIS 2022)* (pp. 35–47). Springer. [doi:10.1007/978-3-031-14314-4_4](https://doi.org/10.1007/978-3-031-14314-4_4)

BOOKS AND CHAPTERS

1. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., Kumar Singh, V., & Sanyal, B. (2023, May). Incentives in Surplus Food Distribution for Smart Cities and Beyond: An Activity Aware Solution. In *Advances in engineering and information science toward smart city and beyond* (pp. 75–90). [doi:10.1007/978-3-031-29301-6_4](https://doi.org/10.1007/978-3-031-29301-6_4)

TECHNICAL & LITERARY SKILLS

Programming	➤ Python, Cython, SLURM, UNIX, Linux, SQL, VBA, C, $\text{\LaTeX}$.
Tools & Systems	➤ PARAM Shivay and Kamrupa Supercomputers, VS Code, Arduino IDE, MySQL.
Communication	➤ Reading, writing, and speaking in English, Hindi, Bengali, Sanskrit, and Assamese.

INDUSTRY EXPERIENCE

2011 – 2019 ➤ **Bio-Statistical Programmer, Data Modeler, and Data Analyst.**
Tata Consultancy Services (TCS), Delhi-NCR, India.

1. I have worked for a Danish Pharmaceutical Client* in the Lifesciences domain as a SAS Bio-statistical programmer doing data analysis and interpretation to establish the safety and efficacy of upcoming drugs undergoing clinical trials in phases 1, 2, and 3.
2. I have also worked for an American Insurance Client* in the Healthcare industry for health-related data modeling and trend prediction of US citizens.

* *Note:* Names of the client organizations have not been disclosed due to contractual obligations.

TEACHING ASSISTANCE & MENTORSHIP EXPERIENCE

TEACHING ASSISTANCE

- IIT Guwahati ➤ **Relational Database Management Systems** for **Coursera** from *September to November, 2026, 2025, 2024.*
- **Cloud Computing** for **Coursera** from *September to November, 2026.*
- **Database Management Systems** for the **Institute** from *January to May, 2026.*
- **Internet of Things** for the **Institute** from *July to November, 2024, 2023, 2022.*
- **Computer Systems** for the **Institute** from *January to May, 2024.*
- **Computer and Network Security** for the **Institute** from *January to May, 2023.*
- **Computer Networks Lab** for the **Institute** from *July to November, 2022.*
- **Topics in Networks** for the **Institute** from *January to May, 2022.*
- IIT (BHU) Varanasi ➤ **Computer Networks** for the **Institute** from *July to November, 2025.*
- **Algorithms** for the **Institute** from *January to May, 2025.*

MENTORSHIP

- IIT Guwahati ➤ **One-week workshop on Power Electronics Hardware Design with Microcontroller Programming** from *June 24 to June 30, 2023* sponsored by the *Science and Engineering Research Board (SERB), India* under the *Accelerate Vigyan* scheme.

CERTIFICATIONS AND TRAINING

- High Performance Computing ➤ **Two-day workshop on High Performance Computing (HPC)** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *Centre for Development of Advanced Computing (C-DAC), India* on *December 2022.*
- Blockchain Applications ➤ **Three-day workshop on Scaling Blockchains: From Bitcoin to the Lightning Network** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *University of Catania, Italy* on *August 2022.*
- Certified Sanskrit Linguist ➤ **Diploma Degree** awarded by the *Central Sanskrit University (CSU), Delhi, India*, organized by the *Indian Knowledge Systems (IKS)* in association with the *Indian Institute of Technology Guwahati, India*, on *June 2024.*
- Certified Karate Trainer ➤ **Second Dan Black Belt and Registered Trainer** awarded by the *Traditional Shito-Ryu Karate Do Association (India), TSKA*, approved by the *Karate India Organization (KIO)*, affiliated with the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC)*, on *December 2022.*

AWARDS AND ACHIEVEMENTS

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| Track and Field | ▶ Active participation awards in Several competitive and charitable track and field events from 2011 to 2020. |
| TCS Delhi-NCR | ▶ Received Service and Commitment award on 2019, 2016, and 2014. |
| | ▶ Received Outstanding Contribution award on 2017, 2016, and 2015. |
| | ▶ Received Distinguished Training award on 2015, 2014, and 2013. |
| | ▶ Received Best Team of the Year award on 2011. |

REFERENCES

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Indian Institute of Technology Guwahati,
Guwahati, Assam, India.

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